SmartPoli Digital Twin: data flow

Team Nexons · GL Bajaj Institute of Technology and Management

1 Data

Patient datasets for training and demo

ShanghaiT2DM (real)
100 patients, glucose every 15 min, meals, doses, health facts
REAL

Demo patient A
real recording from the same dataset
REAL

Demo patient B
CGMacros recording, display only, never trained or scored
DISPLAY ONLY

2 Offline ML (ml/)

Runs once to train and save models

labels.py
low, spike, large rise

features.py
22 inputs, only the past

split.py
79 development patients, 21 locked test patients (split by patient)

XGBoost training
one final test, 16 saved model files

saved models

3 Live inputs per patient

Real-time data used in the app

Static facts
age, BMI, HbA1c, kidney, drugs, years with diabetes

Dynamic stream
glucose every 15 min, meals, insulin and tablet doses

4 Backend (FastAPI)

Inference, alerts and API endpoints (in the app)

twin_features.py
the same 22 inputs rebuilt live

XGBoost forecast
+15, +30, +60, +120 min with 10-90% range

XGBoost warnings
spike above 180 and large rise within 2 h

Low alert
lowest forecast in the next hour vs threshold (90 mg/dL, 100 with heart disease)

Endpoints
/patients/{id}/ml, /twin/worklist, /ml/actions

5 Doctor dashboard (web app)

For clinicians to review and take action

Digital Twin tab
worklist, 2-hour strip, reasons, curve

Doctor actions
note, mark reviewed, snooze

6 Other

Logging and history

timeline

AuditLog
patient timeline and report

7 Hosting

Single deployment with UI, API and models

One Render web service (Docker)
app, API and models together